Chapter 1 – Introduction to Wazuh

1.1 What is Wazuh?

Wazuh is a **free, open-source security platform** that helps organizations **monitor, detect, analyze, and respond to cybersecurity threats**.

It combines the capabilities of:

- SIEM (Security Information and Event Management)
- XDR (Extended Detection and Response)
- File Integrity Monitoring (FIM)
- Vulnerability Detection
- Security Configuration Assessment
- Log Analysis
- Compliance Monitoring

Unlike many commercial SIEM solutions, Wazuh is completely open source and can be installed on your own infrastructure without licensing costs.

1.2 Why Was Wazuh Created?

Modern organizations generate millions of logs every day from:

- Windows Computers
- Linux Servers
- Firewalls
- Routers
- Databases
- Cloud Platforms
- Applications
- VPN Servers

- Active Directory

If every device stores its own logs, security analysts would need to manually review each system during an incident.

This is inefficient and makes investigations difficult.

Wazuh solves this problem by collecting logs from multiple devices, analyzing them in one place, detecting suspicious activities, and generating alerts.

1.3 Why Do Organizations Use Wazuh?

Organizations use Wazuh to:

- Centralize security logs
- Detect cyber attacks
- Monitor endpoints
- Detect malware activity
- Monitor critical files
- Identify vulnerabilities
- Support incident response
- Meet compliance requirements
- Improve overall security visibility

Instead of checking dozens of systems separately, analysts can investigate security events from a single dashboard.

1.4 What Problems Does Wazuh Solve?

Without Wazuh

Windows Logs , Linux Logs , Firewall Logs , Database Logs , Cloud Logs , Separate Investigation

Every system must be checked individually.

With Wazuh

Windows→Linux→Firewall→Cloud→Applications→Wazuh→Correlation→Alert →SOC Analyst

Everything is centralized.

1.5 Main Features of Wazuh

Log Collection

Collects logs from Windows, Linux, macOS, applications, cloud platforms, and network devices.

Log Analysis

Analyzes logs using built-in rules to identify suspicious behavior.

Threat Detection

Detects activities such as:

- Brute-force attacks
- Privilege escalation
- Malware execution
- Unauthorized file changes
- Suspicious PowerShell activity

File Integrity Monitoring (FIM)

Monitors important files and alerts when they are:

- Created
- Modified
- Deleted
- Renamed
- Tampered with

Vulnerability Detection

Detects outdated software and known vulnerabilities by comparing installed software against vulnerability databases.

Security Configuration Assessment (SCA)

Checks whether operating systems follow recommended security configurations.

Examples:

- Password policy
- Firewall configuration
- SSH configuration
- Account settings

Compliance Monitoring

Helps organizations meet security standards and regulations such as:

- PCI DSS
- HIPAA
- GDPR
- CIS Benchmarks

Dashboard

Provides an easy-to-use web interface where analysts can:

- Search logs
- View alerts
- Investigate incidents
- Monitor endpoints
- Review vulnerabilities

1.6 Wazuh Components

A complete Wazuh installation contains four major components.

1. Wazuh Agent

Installed on endpoints such as:

- Windows
- Linux
- macOS

Responsibilities:

- Collect logs
- Monitor files
- Detect local events
- Send information to the Wazuh Manager

Think of the agent as the **eyes and ears** on each computer.

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2. Wazuh Manager

The Manager is the **brain** of Wazuh.

Responsibilities:

- Receive data from agents
- Analyze logs
- Apply detection rules
- Generate alerts
- Manage connected agents

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3. Wazuh Indexer

Stores all collected logs and alerts.

Responsibilities:

- Index data
- Store security events
- Enable fast searching

Without the Indexer, searching millions of logs would be very slow.

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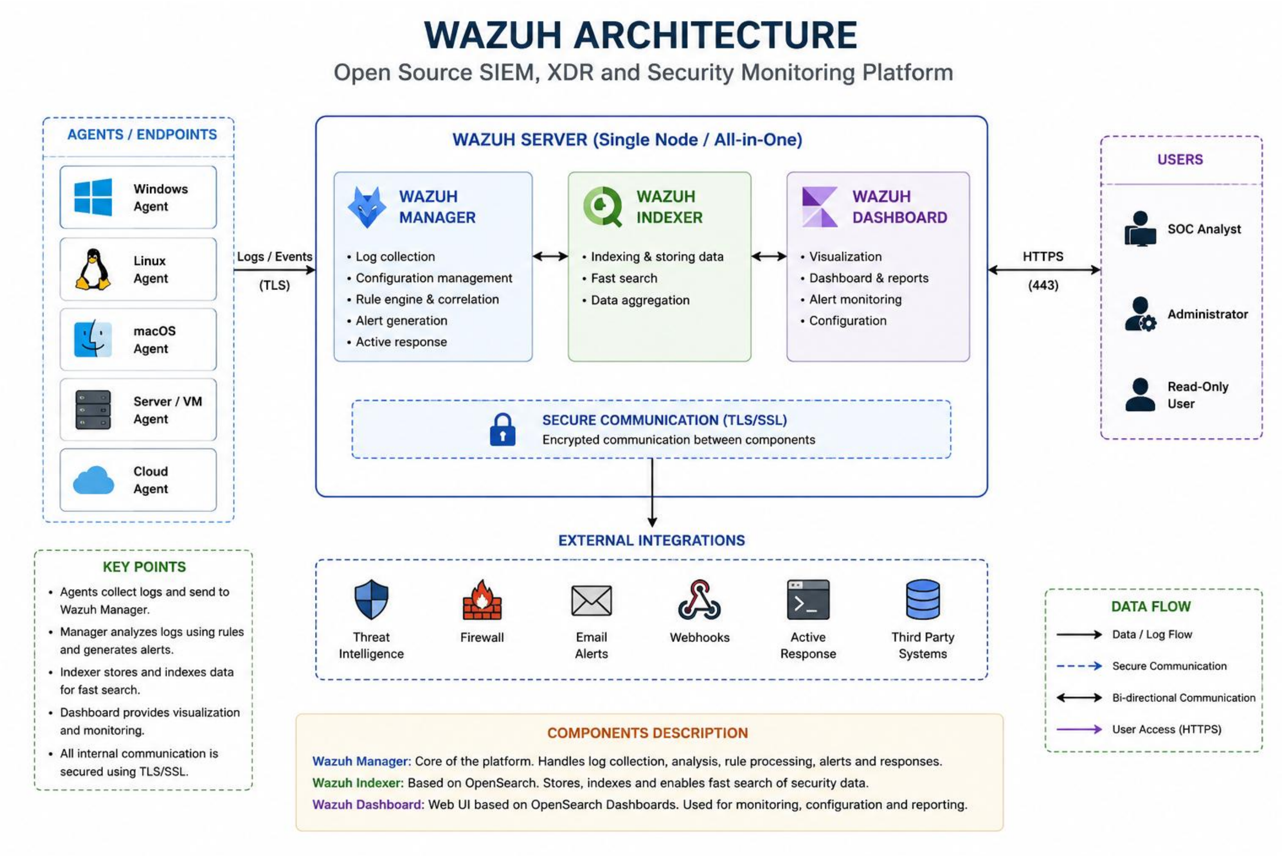
4. Wazuh Dashboard

The Dashboard is the graphical web interface used by SOC analysts.

Responsibilities:

- Display alerts
- Search logs
- View dashboards
- Investigate incidents
- Manage agents

1.7 Wazuh Architecture



1.8 How Wazuh Works

Step 1

An endpoint generates an event.

Example:

A user fails to log in.

Step 2

The Wazuh Agent collects the event.

Step 3

The Agent sends the event securely to the Wazuh Manager.

Step 4

The Manager analyzes the event using detection rules.

Step 5

If suspicious activity is detected, an alert is generated.

Step 6

The alert is stored in the Wazuh Indexer.

Step 7

The SOC Analyst views the alert in the Wazuh Dashboard.

1.9 Real-World Example

Imagine an attacker attempts to brute-force a Windows machine.

The process looks like this:

Attacker→Windows Login Attempts→Windows Event Log→Wazuh Agent→Wazuh Manager→Detection Rule→High Severity Alert→Dashboard→SOC Analyst Investigation

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1.10 Why Wazuh is Popular

Advantages:

- Completely Free
- Open Source
- Enterprise Features
- Active Community
- Supports Windows, Linux, and macOS
- Built-in File Integrity Monitoring
- Built-in Vulnerability Detection
- MITRE ATT&CK Integration
- Excellent for SOC Learning

1.11 Wazuh vs Splunk

Wazuh	Splunk
Free & Open Source	Commercial (Paid)
SIEM + XDR	Primarily SIEM Platform
Built-in Endpoint Monitoring	Requires additional integrations for some capabilities
Excellent for Home Labs	Widely used in large enterprises
Community Driven	Commercial Vendor Support

1.12 Key Terms

SIEM: Collects, stores, analyzes, and correlates logs.

XDR: Detects and responds to threats across multiple security sources.

EDR: Detects and responds to threats on endpoint devices.

FIM: Monitors important files for unauthorized changes.

Agent: Software installed on an endpoint that collects security events.

Manager: Central server that analyzes logs and generates alerts.

Indexer: Stores and indexes logs for fast searching.

Dashboard: Web interface used to monitor and investigate security events.

Chapter 1 Summary

- Wazuh is a **free, open-source SIEM and XDR platform**.
- It helps organizations **collect, analyze, correlate, and monitor security events** from multiple systems.
- Wazuh consists of four main components:

    - Agent
    - Manager
    - Indexer
    - Dashboard
- It supports features such as:
    - Log Analysis
    - Threat Detection
    - File Integrity Monitoring (FIM)
    - Vulnerability Detection
    - Compliance Monitoring
- Wazuh is widely used for **SOC operations, security monitoring, incident response, and cybersecurity home labs**.

One-Line Summary

Wazuh is a free, open-source SIEM and XDR platform that collects logs from endpoints, analyzes them for suspicious activity, stores them for investigation, and presents alerts through a centralized dashboard to help security teams detect and respond to cyber threats.